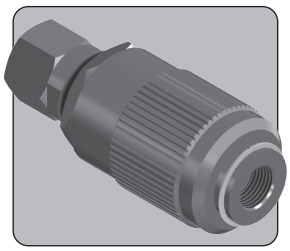
IMC50

METERING VALVE

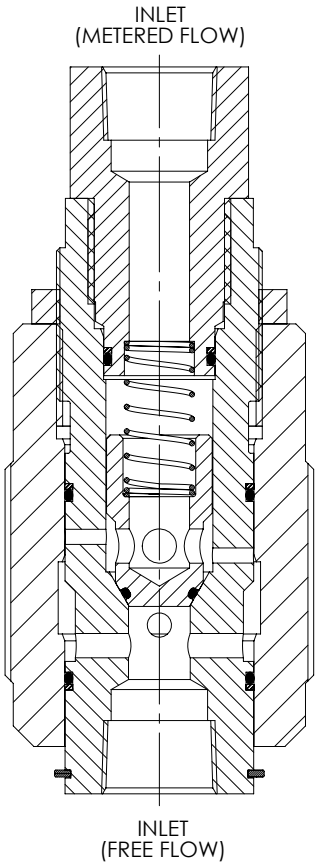
1/2″ UP TO 552 BAR 8,000 PSI



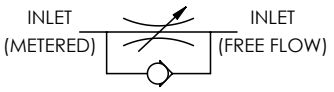
The IMC50 is a 1/2″ nominal bore metering valve with a reverse free-flow check function. It provides metered flow in one direction, and free flow in the opposite direction. It is ideal for uni-directional speed control of a cylinder or actuator. The valve may also be referred to as a flow control valve.

Note that this valve does not provide tight shut-off.

- Metering valve with reverse free-flow check feature
- Inline design with barrel style adjustment offers maximum space efficiency
- Lockable adjustment
- Ideal for uni-directional speed control
- Stainless steel construction
- Variants can be tailor-made to suit specific metering characteristics. Contact us for advice
- Suitable for use with mineral oils, water glycols and plain water
- Suitable for many other media, contact us for advice
- Various porting options available
- Various seal options available



Specifications

BASIC MODEL NUMBER	IMC50
SYMBOL	
MAX WORKING PRESSURE (LIQUID)	552 bar (8,000 psi)
CV (FLOW CAPACITY)	Up to 1.2 (Metered) 3.0 (Free flow) See Typical Performance Graph
FLUID	Liquids only See materials section
TEMPERATURE RANGE	See Product Selector opposite and Technical Data section
PORT SIZE	1/2″
CHECK ELEMENT CRACKING PRESSURE	0.1 - 0.3 bar (2 - 5 psi)
WEIGHT	1.25 kg (2.8 lb)

Specifications may change without notice

Materials

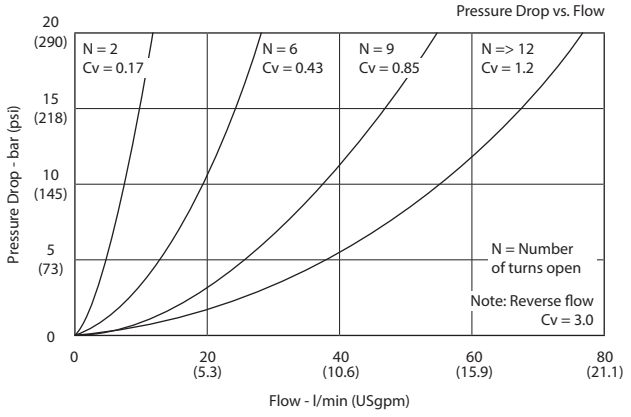
Externally Exposed Parts: 316 Stainless Steel with a 302 Series Stainless Steel adjustment stop circlip.

Internally Wetted Parts: 316 and 302 Stainless Steel and Acetal.

The standard valve is designed for use with mineral oils, water glycols, plain water and may also be used with a wide variety of media compatible with the materials of construction. Please contact us for advice when different construction materials are required.

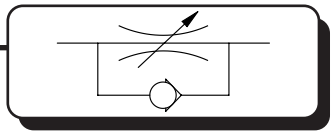
⚠ The standard valve has Viton® seals. Further seal options are available via the Product Selector. Compatibility with the working fluid at the operating temperature must be considered.

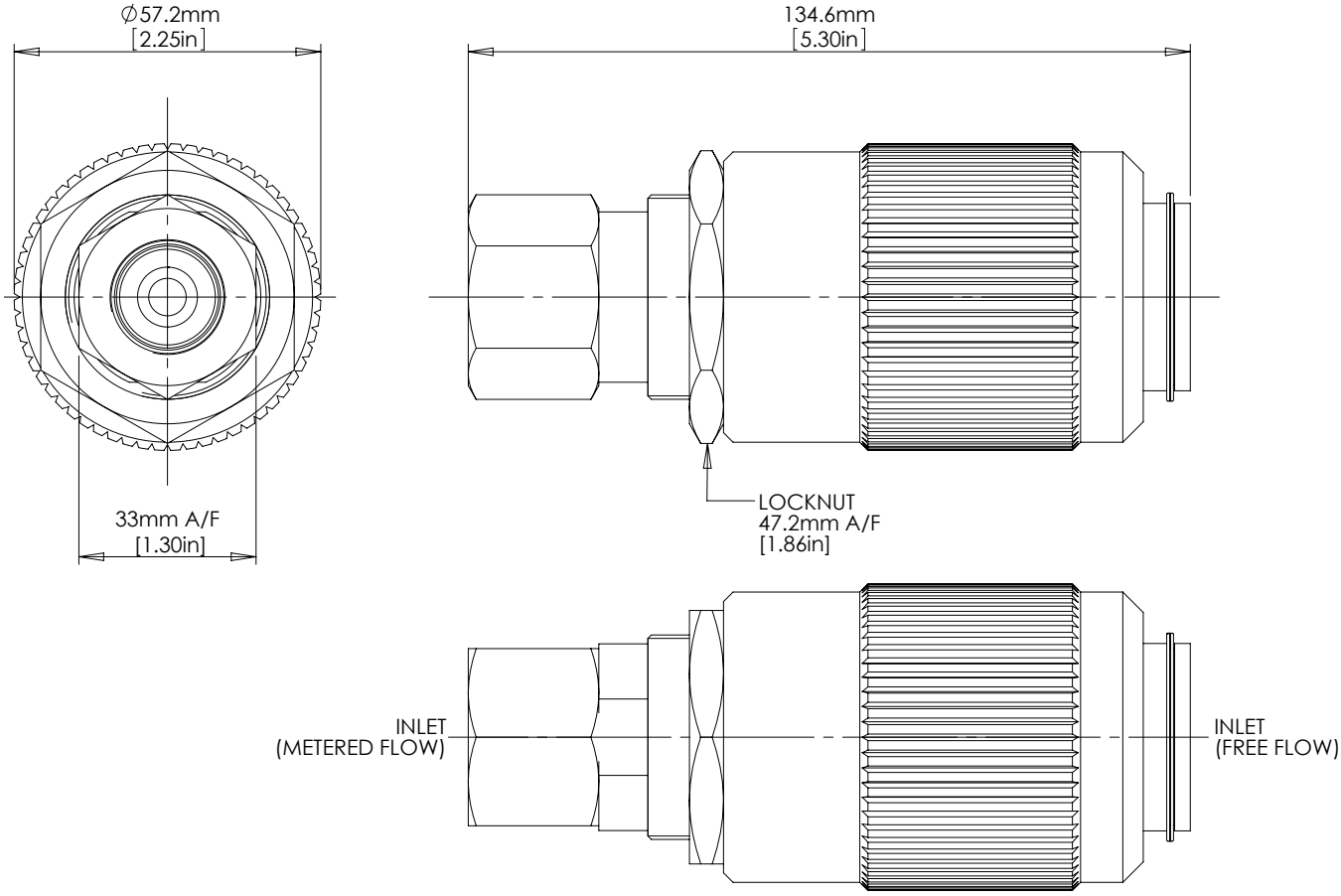
Typical Performance



Typical performance based on water

Installation Diagram





Product Selector

IMC50 P - NBR - 8K

VALVE TYPE	PORTING OPTIONS	SEAL OPTIONS	MAX WORKING PRESSURE
IMC50	P 1/2" BSPP female N 1/2" NPT female	Leave blank for Viton® seals -10˚C to +120˚C NBR Nitrile seals -30˚C to +105˚C EP EPDM seals -45˚C to +120˚C KLZ Kalrez® seals -25˚C to +120˚C HNBR Hydrogenated Nitrile seals -20˚C to +120˚C Further seal options are available on request ⚠ Exposure to high and low fluid temperatures will influence handle temperature. Operator safety must be considered.	8K 552 bar (8,000 psi) Liquid

Repair and seal kits are available on request

For a seal kit add SK at the end of the model code and for a repair kit add RK